

Peptide-MHC Binding Prediction with Deep Learning

Manuscript Data Package

Date: June 15, 2026

Project: Deep learning for HLA-A*02:01 epitope prediction

Model: Deep FFN (360-180-90-45-3), MHCflurry 2.2.0 trained

Performance: 91.9% accuracy, 89.6% CV, AUC 0.947, 93.9% sensitivity

Data Inventory

Core Datasets

File	Rows	Columns	Description
data/real_peptide.csv	5,088	8	MHCflurry-labeled training data
data/feature.csv	5,088	282	FFN features (BLOSUM + learned + predictions)
data/feature_resnet.csv	5,088	315	ResNet features
data/model_comparison.csv		6	All models – accuracy + per-class F1
data/cv_summary.csv		2	5-fold CV fold-by-fold

Epitope Prediction Results

File	Rows	Description
data/protein_epitope_scan.csv	772	3 proteins (MART-1, Spike RBD, p53)
data/protein_epitope_scan_extended.csv	2,764	7 additional proteins
data/top20_epitope_candidates.csv	20	Ranked top candidates
data/new_peptide_predictions.csv		Test peptide predictions
data/iedb_benchmark_results.csv	60	49 validated epitopes + 20 controls
data/mutation_scan_results.csv	144	p53/KRAS hotspot neoantigens

Trained Models

File	Size	Architecture	Accuracy
models/FFN_DeepMB	4.1 MB	360-180-90-45-3	91.9%
models/FFN_JessieK20	4.2 MB	180-90-3	90.9%
models/CNN_JessieK20	8.5 MB	Conv2D + FFN	90.0%

File	Size	Architecture	Accuracy
models/LSTM_E2E_KD.h5	328-KB	LSTM-64 + Dense	83.3%
models/ResNet151B.h5	1.5-GB	3-block residual	84.3%
models/RandomForest.rds	1-MB	100 trees	81.1%
models/cv_FFN20KFold_{1.5-50}.h5	120-KB	50 1.5-fold CV ensemble each	89.6%

Figures

File	Content
plots/comparison_accuracy.png	BSSM vs MHCflurry model accuracy
plots/comparison_per_class_f1.png	DeepFFN per-class F1
plots/comparison_cv_folds.png	5-fold CV comparison
plots/comparison_gap.png	Data source accuracy gap
plots/benchmark_roc.png	ROC curve, AUC 0.947
plots/benchmark_confusion_matrix.png	Confusion matrix heatmap
plots/benchmark_per_epitope.png	Per-peptide prediction scores
plots/feature_blosum_images.png	BLOSUM62 encoding (HOT)
plots/feature_learned_heatmap.png	Learned feature representations
plots/feature_clustered_heatmap.png	Clustered feature heatmap
plots/feature_correlation.png	Feature-binding score correlation
plots/mutation_delta_scores.png	Mutation delta score bar chart
plots/mutation_wt_vs_mutant.png	WT vs mutant paired plot
plots/mutation_protein_map.png	WT/KRAS mutation map

Reports (Markdown + PDF)

File	Content
literature-map-epitope-prediction.*	90+ paper literature review
project_summary.*	Full project overview
study-design.*	Methodology and research questions
protein_epitope_scan.*	First 3-protein scan
extended_scan_report.*	10-protein consolidated scan
top20_epitope_candidates.*	Ranked top 20 with annotations
prediction_results.*	10 test peptide predictions
iedb_benchmark_report.*	49-epitope benchmark
mutation_scan_report.*	Cancer hotspot neoantigens

Source Code

File	Description
peptide_mhc_binding_study.R	Main pipeline (1,700+ lines)
scan_protein_epitopes.R	Protein epitope scanner
scan_more_proteins.R	Extended protein scanner
predict_new_peptides.R	New peptide prediction
benchmark_iedb_epitopes.R	IEDB benchmark
mutation_scan.R	Cancer mutation scanner
plot_comparison.R	Comparison plots
plot_feature_maps.R	Feature visualization
plot_benchmark_roc.R	ROC + confusion matrix
plot_mutation_delta.R	Mutation delta plots
label_peptides_mhcflurry.py	MHCflurry peptide labeling
run_main_pipeline.R	Pipeline runner

Performance Summary

Model Benchmarks

Metric	Value
Best model accuracy	91.9%
5-fold CV mean	89.6% +/- 0.8%
IEDB sensitivity	93.9% (46/49)
IEDB specificity	75.0%
ROC AUC	0.947
F1 score	0.920

Epitope Discovery

Metric	Value
Proteins scanned	10
Total 9-mers	3,536
Strong binders found	96 (2.7%)
Known epitopes confirmed	10/11 (91%)
Novel candidates (score=1.0)	12
Neopeptides discovered	2 (p53 R248W, KRAS G12V)

Data Source Comparison

Label Source	Best Accuracy	WB F1
MHCflurry 2.2.0	91.9%	0.880
PSSM (biophysics)	94.8%	0.925
Random synthetic	65.8%	0.000

Citation

If you use this data in a publication, please cite:

1. Jessen, L.E. (2018). “Deep Learning for Cancer Immunotherapy.” RStudio AI Blog.
2. O’Donnell, T.J. et al. (2020). “MHCflurry 2.0.” Cell Systems, 11(1), 42-48.
3. Reynisson, B. et al. (2020). “NetMHCpan-4.1.” Nucleic Acids Research, 48(W1).
4. This work: “Deep learning architectures for peptide-MHC class I binding prediction.” (in preparation)

File Manifest

```

Peptide epitope/
|-- data/
|   |-- real_peptides.csv           5,088 training peptides
|   |-- feature.csv                5,088 x 282 FFN features
|   |-- feature_resnet.csv         5,088 x 315 ResNet features
|   |-- model_comparison.csv       6-model comparison
|   |-- cv_summary.csv            5-fold CV results
|   |-- protein_epitope_scan.csv   772 peptides (3 proteins)
|   |-- protein_epitope_scan_extended.csv 2,764 peptides (7 proteins)
|   |-- top20_epitope_candidates.csv Top 20 ranked
|   |-- new_peptide_predictions.csv 10 test predictions
|   |-- iedb_benchmark_results.csv 69-peptide benchmark
|   |-- mutation_scan_results.csv  144 mutation windows
|
|-- models/
|   |-- FFN_Deep.h5                Best model (91.9%)
|   |-- FFN_Jessen2018.h5          Jessen baseline
|   |-- CNN_Jessen2018.h5          CNN model
|   |-- LSTM_Extended.h5           LSTM model
|   |-- ResNet_Style.h5            ResNet model
|   |-- Random_Forest.rds          RF baseline
|   |-- cv_FFN_CV_Fold_{1..5}.h5  5-fold CV ensemble
|   |-- model_manifest.csv         Model index

```

-- plots/	14 figures (PNG)
-- *.R	11 R scripts (reproducible)
-- *.py	2 Python labeling scripts
-- *.md + *.pdf	8 report pairs
-- MANUSCRIPT_README.md	This file

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